

**Smart Fisher Lanka is an AI-based decision support
system designed for Sri Lankan fishermen**

25-26J-177

Project Proposal Report

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B.Sc. (Hons) Degree in Information Technology specialized in
Information Technology

Department of Information Technology

Sri Lanka Institute of Information Technology

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Research Component: Fishing Trip Cost Estimation and Management

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DECLARATION

We declare that this is our own work, and this proposal does not incorporate without acknowledgement any material previously submitted for a degree or diploma in any other university or Institute of higher learning and to the best of our knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgement is made in the text.

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The above candidate is carrying out research for the undergraduate Dissertation under my supervision.



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ABSTRACT

Small scale fisheries play an important role in supporting the livelihoods and nutrition of Sri Lanka's coastal communities. Although many artisanal fishers still depend on guess works instead of using structured, data-driven methods when planning their fishing trips. This makes a clear gap in knowledge, as current practices often overlook important factors like fuel price changes, route distances, gear readiness, crew size, catch uncertainty, and regulatory requirements. As a result, fishers often lose money, face safety problems, and struggle to follow fishing rules.

Current global platforms, such as FishTrack, PeskaAAS, Abalobi, and Deckhand, mainly target large-scale, Western fisheries. They are costly, language-limited, lack offline functionality, and do not fit the social realities of Sri Lankan artisanal fishers. This highlights the novelty of the proposed work: a localized, offline-first, and fisher-friendly decision support system with fish-zone-based cost prediction, integrated inventory management, and automated license expiry alerts.

The mobile-based system combines three specialized domains: Data Science (to process historical and real-time inputs such as fuel prices, labor, and market rates), Geospatial Analysis (to calculate costs dynamically based on fishing zone distances), and Human-Computer Interaction (HCI) (to provide a simple, Sinhala/Tamil offline-ready interface). Its architecture includes three layers: a data collection layer capturing fuel, labor, gear, inventory, and fish zone data; a Python-based calculation engine for dynamic trip cost estimation with safety buffers and penalties; and a

presentation layer with a user-friendly mobile UI offering trip simulations, inventory reminders, and license alerts.

The system will use React Native, SQLite, and RESTful APIs for offline data management and external integrations. Evaluation benchmarks include reducing trip planning time by 20%, achieving 90%+ accuracy in cost predictions, and gaining 90%+ user satisfaction. Development will follow an agile 5-month work breakdown structure (WBS) covering requirements, UI/UX prototyping, algorithm development, integration, testing, and deployment.

By introducing zone-based cost prediction, localized inventory management, and compliance alerts, the system bridges a critical gap in Sri Lanka's artisanal fisheries, improving decision-making, safety, and long-term sustainability, with potential adaptation for other developing nations.

Keywords: Decision Support System, Fish-Zone-Based Cost Estimation, Inventory Management, Small-Scale Fisheries, Sustainable Livelihoods, Mobile Application, Localization, Sri Lanka, Data Science, Geospatial Analysis

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1. INTRODUCTION

Small-scale fisheries (SSF) are not only about catching fish, but they are also important for food security, employment, and the stability of coastal communities. In Sri Lanka, they play an important role in daily nutrition and income generation. Yet, every fishing trip carries a high degree of financial uncertainty. Fishers must consider costs such as fuel, ice, crew wages, and gear, while estimating the income from their potential catch. Most of these decisions are still based on personal experience, traditional knowledge or guess works rather than structured data. With fuel prices increasing and market prices fluctuating. So, trips that seem profitable at the start of the day can easily result in losses [1], [2].

Technology is gradually reshaping this situation. In agriculture, decision-support systems are already used to improve yield estimates, manage costs, and support better planning [3]. In fisheries, digital tools such as PeskAAS and Abalobi have created new ways for fishers to report catches, record sales, and link directly with buyers [4], [5]. However, these systems mostly provide insights after the trip, offering little support for pre-trip planning.

Recent advances in mobile applications, GPS, and data analytics are opening new opportunities for small-scale fisheries. GPS and mapping technologies can optimize routes, while APIs provide real-time information on weather and fuel prices [6]. Machine learning also enables predictions based on historical fishing data, such as estimating trip costs, identifying productive fishing zones, and reducing unnecessary expenses [7]. These tools allow fishers to move from guesswork to evidence-based decisions, which can increase both safety and profitability.

Current fisheries applications remain limited, as they often focus only on catch reporting or marketing [4], [5]. Few integrate financial forecasting, inventory readiness, and compliance alerts in one platform. The proposed Trip Cost Estimation and Management System aims to address this gap by combining

location-based cost models, fuel and weather updates, and inventory checks into a single, offline-ready mobile solution. By doing so, it offers a localized, practical, and scalable approach for Sri Lankan fishers, helping them plan safer and more profitable trips before leaving shore

2. Background & Literature survey

Small-scale fisheries (SSF) represent far more than an economic activity; they are the vibrant, beating heart of coastal communities across the globe, contributing significantly to global food security and employment [1]. In Sri Lanka, this sector is a critical linchpin, directly underpinning daily nutrition for millions and providing a primary source of income for over 100,000 fishers and their families [2]. However, this lifeline is perpetually frayed by profound uncertainty. Each voyage to sea is a high-stakes gamble against a volatile ocean and an unpredictable market. Fishers must make crucial pre-trip decisions—estimating potential income against costs like fuel, ice, crew shares, and gear maintenance—often relying on generations of traditional knowledge and intuition rather than empirical, data-driven insight. In an era of soaring fuel prices [2] and fluctuating fish market values, this reliance on guesswork can swiftly turn an anticipated profitable day into a significant financial loss, jeopardizing the well-being of entire households [2], [8].

The transformative potential of technology to address such challenges is already evident in parallel sectors. In agriculture, the advent of Agriculture 4.0 has seen digital decision-support systems revolutionize farming practices, enabling precise yield forecasts, optimized resource allocation, and enhanced financial planning through IoT and data analytics [3]. This success story provides a powerful template for the fisheries sector. The digital revolution has begun to reach the world's coasts with platforms like PeskAAS, which offers near-real-time catch monitoring and data visualization for fisheries management [4], and Abalobi, which connects fishers to markets and enhances traceability, empowering them within the

value chain [5]. These tools have made significant strides in improving post-harvest transparency and value. However, a critical gap remains: they offer limited support for the precarious planning phase before the boat even leaves the shore, leaving fishers to navigate their most financially risky decisions without modern tools.

Fortunately, recent technological advancements are creating unprecedented opportunities to bridge this gap. The ubiquitous penetration of mobile phones even in remote coastal areas provides a delivery mechanism for powerful tools. Geospatial technologies and APIs, such as the Google Distance Matrix API [6], can calculate optimal navigation routes, directly translating saved kilometers into saved fuel and money. Other APIs can pull in real-time, hyper-local data on weather patterns and daily fuel prices, factors that are notoriously variable and central to a trip's viability. Furthermore, artificial intelligence and machine learning techniques can analyze historical catch and environmental data to identify patterns, predict potential yields in specific fishing zones, and generate accurate, personalized cost estimates, empowering fishers to make evidence-based decisions [7], [9].

Beyond fuel and catch, two other pillars of operational readiness are often overlooked: inventory and compliance. Research indicates that trips are frequently delayed or incur unexpected costs due to missing, damaged, or expired gear, from nets to critical safety equipment [10]. An integrated system that tracks inventory readiness is not merely an organizational tool; it is a direct financial safeguard. Similarly, regulatory compliance,

particularly regarding fishing license validity as outlined by national authorities [11], is a common pain point. Overlooked expiry dates can result in heavy fines or confiscated catch, devastating a fisher's income. An automated alert system for such deadlines is a simple yet profoundly effective feature for risk reduction.

The current landscape of fisheries applications is fragmented. Most solutions are siloed, focusing either on catch reporting (like PeskAAS [4]) or on market access (like Abalobi [5]). There is a conspicuous absence of a holistic, integrated platform that combines pre-trip financial forecasting, inventory management, and regulatory compliance into a single, user-friendly application designed for the specific constraints of small-scale fishers—namely, low cost and offline functionality.

It is precisely this gap that the proposed Trip Cost Estimation and Management System aims to fill. This research proposes the development of a comprehensive mobile solution that synthesizes the above elements. By combining a location-based cost calculation engine, real-time data feeds, inventory checks, and compliance alerts into an offline-ready application, the system offers a localized and practical tool for Sri Lankan fishers. The envisioned system architecture and its data flow are designed to be clear and robust, as illustrated in the following diagrams.

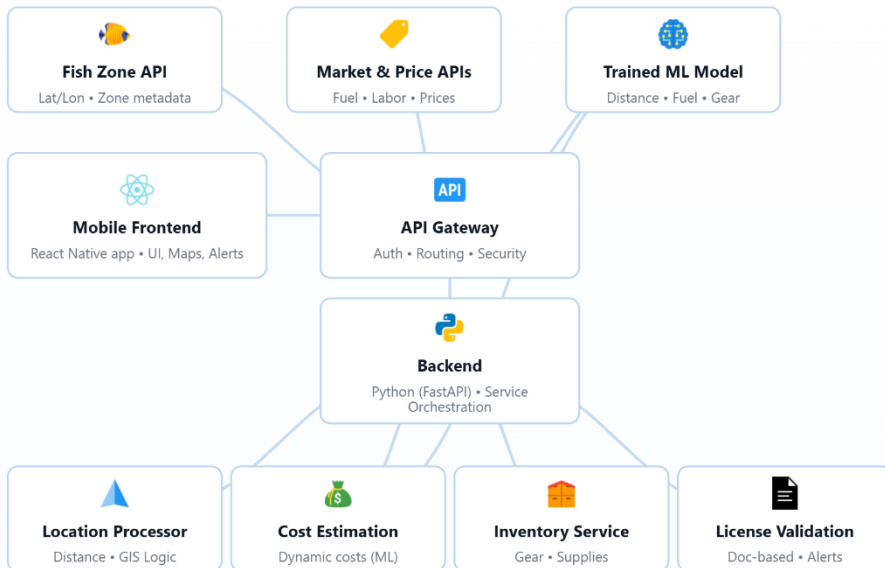


Figure 1.1

Figure 1.1 shows the process where data from multiple sources is processed in a central engine and delivered to the user. The diagram represents a functional architecture of the system, displaying how various services and tools interact to support digital decision-making. The API Gateway is in the middle of architecture and is the primary entry point. There are three key responsibilities of the API Gateway which handle authentication, routing,

and security. - Maintains organization as well as secure connections with other components of architecture

From the top layer

- Fish Zone API provides critical geospatial data such as latitude/longitude and metadata about fishing zones.
- Market & Price APIs supply real-time information on fuel, labor, and gear prices.
- Trained Machine Learning (ML) Model processes multiple inputs (distance, fuel, gear) to make predictive assessments that improve decision-making.
- Mobile Frontend (built with React Native) is the user interface, displaying maps, alerts, and real-time updates directly to end-users.

At the middle layer

- The Backend (developed using Python/FastAPI) orchestrates services and runs the logic behind the system. It integrates with the gateway and ensures smooth coordination of all APIs and models.

At the bottom layer

- Location Processor applies GIS-based logic to calculate distances and routes.
- Cost Estimation uses ML to dynamically compute operational costs (fuel, labor, gear).

- Inventory Service tracks gear, supplies, and other essential resources.
- License Validation ensures compliance by checking documents and sending alerts.

In essence, each block in the diagram represents a **specific function** (data input, processing, analysis, validation, or user interaction). When combined, they create a **complete ecosystem** that improves operational efficiency, reduces risks, and supports smarter decisions

3. Research Gap

Most existing trip planning or cost estimation tools are not designed with fishers and small boat owners in mind. Many available systems are general, focused on large-scale shipping or logistics, and they do not provide real-time data updates that fishers need when planning daily trips. Another gap is that these tools are not localized; they don't consider specific conditions like local fuel prices, regional weather changes, or nearby fishing routes. Because of this, fishers are left depending on guesswork, old experience, or scattered apps that don't talk to each other. This often leads to poor cost estimation, wasted fuel, and financial uncertainty. There is also a lack of integrated platforms that combine fuel, crew wages, inventory, weather, and maps into one simple system. My research addresses this gap by developing a real-time, localized, and user-friendly solution to support smarter and more reliable decision-making for fishers. Fragmentation across functionalities.

Existing tools such as PeskAAS for monitoring catches [4] and Abalobi for market traceability [5] address specific issues but fail to provide a unified, pre-trip decision-support system. Very few solutions combine predictive yield forecasts, cost estimation, inventory readiness, and license validation into a single workflow, forcing fishers to juggle multiple sources of information.

Lack of localized, predictive cost modeling.

While global studies and APIs can estimate distances or provide raw fuel data [6], there is no integrated system that transforms these dynamic inputs into actionable cost forecasts for Sri Lankan fisheries. Fishers still rely on

intuition rather than predictive analytics to balance expenses (fuel, crew, gear) against potential income [1][2].

Absence of inventory-integrated financial planning

Although research highlights the importance of gear readiness and safety equipment [10], no available tool connects equipment status directly to trip feasibility. Missing or damaged gear results in both safety risks and hidden financial losses, yet this remains excluded from most digital solutions.

Limited offline-first and low-tech design..

Applications like Abalobi [5] assume constant connectivity and digital fluency. For rural Sri Lankan fishers operating in remote waters, poor internet access and varying literacy levels make many of these tools impractical. Offline-first architectures with simplified mobile interfaces remain underexplored

Insufficient integration with existing prediction models

Recent advances in ML for predicting fishing grounds [7][9] remain siloed from practical financial planning tools. Current systems do not integrate these predictions into real-time cost-benefit analyses that would allow fishers to simulate trip profitability before departure [3][8]

Research / System	Predictive Cost Estimation	Route/Distance Optimization	Inventory & Gear Tracking	License/ Compliance Alerts	Offline-first, Low-cost Design
PeskAAS [4]	×	×	×	Partial (catch reporting)	×
Abalobi [5]	×	×	×	×	(requires connectivity)
Google Distance API [6]	×	×	×	×	×
Deep learning catch models [7][9]	×	×	×	×	×
Gear readiness studies [10]	×	×	×	×	×
Proposed System	✓	✓	✓	✓	✓

Figure 1.2: Summary of Research and System Gaps in Existing Fisheries Tools

4. RESEARCH PROBLEM

Small-scale fishers in Sri Lanka face significant uncertainty before embarking on fishing trips. Critical operational decisions—such as estimating fuel costs, planning optimal routes, ensuring inventory readiness, and verifying license validity—are primarily based on intuition rather than systematic, data-driven processes. Each trip carries financial and safety risks, and any miscalculation can result in wasted resources, unprofitable trips, or regulatory penalties

Current digital platforms, such as PeskAAS [4] and Abalobi [5], mainly support post-harvest monitoring, market access, or catch reporting. These systems do not provide integrated pre-trip planning tools that combine cost forecasting, inventory tracking, route optimization, and compliance alerts. Most solutions also require continuous internet access and assume a high level of digital literacy, making them less practical for rural coastal communities with limited connectivity

Interviews with local fishers and review of fisheries management reports [1][2][10] reveal that while fishers are willing to adopt technology, there is no existing system that provides predictive cost estimation, integrates inventory readiness, validates licenses, and recommends profitable fishing zones using machine learning—all in an offline-ready, mobile-friendly format. This creates a clear gap: a unified, practical tool for pre-trip decision-making that reduces uncertainty, financial risk, and operational inefficiencies

.

This research therefore identifies a substantial gap in developing a holistic, localized, offline-ready mobile system that supports:

- Trip cost estimation using predictive modeling of fuel, labor, and gear.
- Route optimization through GIS-based distance calculations.
- Inventory and safety checks integrated into financial forecasting.
- Regulatory compliance monitoring with automated alerts.
- Catch prediction using ML models trained on local and environmental data.

The proposed system addresses these challenges by creating a decision-support tool specifically tailored to the socio-economic and technological realities of Sri Lanka's small-scale fisheries sector. It aims to reduce pre-trip uncertainty, minimize risks, and empower fishers with evidence-based planning capabilities

5.OBJECTIVES

5.1 MAIN OBJECTIVES

The main goal of the study is to design and construct a mobile application for small scale fisher's application that provides predictive trip cost estimation; inventory tracking; compliance alerts (regarding their licenses); optimized trip route information; and catch prediction using machine learning. The application will use and integrate data from multiple sources; provide information for fishers to make better decisions; and function properly offline when there is limited connectivity. The application is designed to help fishers to make more evidence-based decisions to increase trip profit while minimizing operational risk..

5.2 SPECIFIC OBJECTIVES

5.2.1. Estimate trip costs based on dynamic variables.

- Calculate projected fuel, labor, and gear expenses before departure.
- Integrate real-time and historical data into machine learning models to simulate trip profitability.

5.2.2. Optimize fishing routes using geospatial analysis.

- Use GIS and location-based APIs to suggest efficient routes to fishing zones.
- Reduce fuel consumption and travel time while considering safety and distance constraints

5.2.3 Integrate inventory and safety checks into pre-trip planning

- Track gear, ice, and safety equipment readiness.
- Alert fishers about missing, damaged, or expired items, reducing risk and unforeseen costs

5.2.4. Monitor regulatory compliance.

- Validate licenses and permits automatically.
- Send alerts for expiry dates to prevent fines and confiscation of catch

5.2.5. Enable offline-first operation and low-tech usability.

- Ensure core functionalities work without internet access.
- Provide a simple, mobile-friendly interface suitable for users with varying levels of digital literacy.

6. METHODOLOGY

6.1. Understanding the Research Domain

6.1.1 Machine Learning for Predictive Analytics

The proposed system uses machine learning to predict fishing trip costs and expected catch by analyzing historical trip records, fish zone coordinates (longitude and latitude), gear condition, fuel consumption, crew wages, and environmental factors such as weather and sea conditions. Supervised learning algorithms like Random Forest or Gradient Boosting generate data-driven estimates of potential revenue and operational costs and helping fishers to plan trips more efficiently. As illustrated in the figure 1.2 the system is organized in layers which the **data layer** collects inputs from external APIs and local databases. Also the **preprocessing and feature engineering layer** cleans data and extracts key features like distance, gear wear, and predicted catch scores; the **model training layer** develops predictive models. Then the **inference and backend layer** provides real-time predictions through Flask/FastAPI endpoints. After that the **business logic layer** applies rules such as safety buffers and license expiry alerts; and the **presentation layer** in the React Native app displays results, enabling real-time, location-aware decision-making and inventory management for fishers

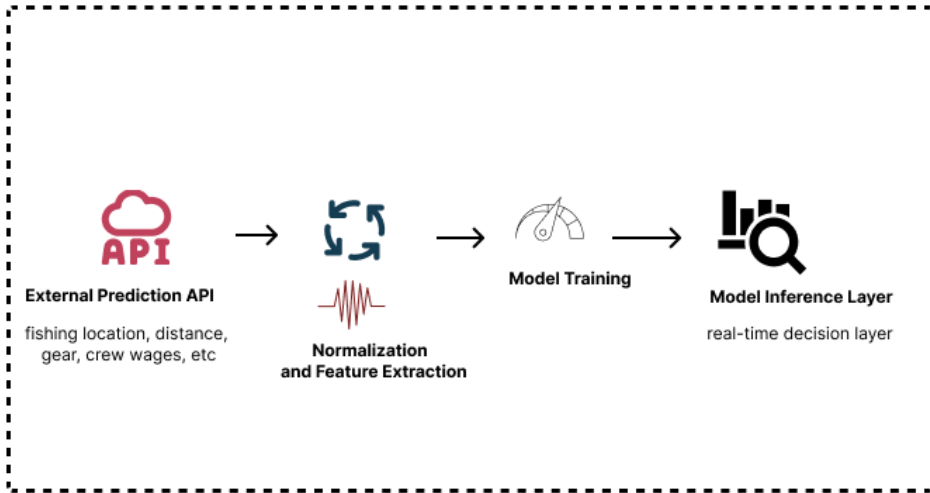


Figure 1.2

6.1.2 Geospatial Analysis and Route Optimization

Geospatial analysis involves processing location-based data to derive meaningful insights. For this system, APIs like the Google Distance Matrix API are employed to calculate the precise distance between the fisher's launch point and their chosen fishing zone. This calculated distance is a fundamental input for the fuel cost estimation module. Route optimization algorithms consider about factors such as shortest path and potential weather barriers, directly contributing to fuel savings and trip time reduction, which are crucial for profitability.

6.1.3 Microservices Backend Architecture

The backend will be developed as a microservices app utilizing FastAPI. This involves structuring the app as a set of loosely coupled fine grained different services, (e.g. cost estimating service, inventory service, weather service), built using the RESTful API framework. This allows for greater flexibility, scalability, and maintainability, and because each service can be deployed and scaled independently, it is well suited for bringing together many aspects and data sources in a single system.

6.1.4 Microservices Backend Architecture

The backend system is designed around a microservices architecture using FastAPI. This approach structures the application as a collection of loosely coupled, fine-grained services (e.g., a cost estimation service, an inventory service, a weather service) that communicate via RESTful APIs. This enhances adaptability, scalability, and maintainability. Each service can be developed, deployed, and scaled independently, which is ideal for integrating diverse functionalities and data sources into a single, cohesive system.

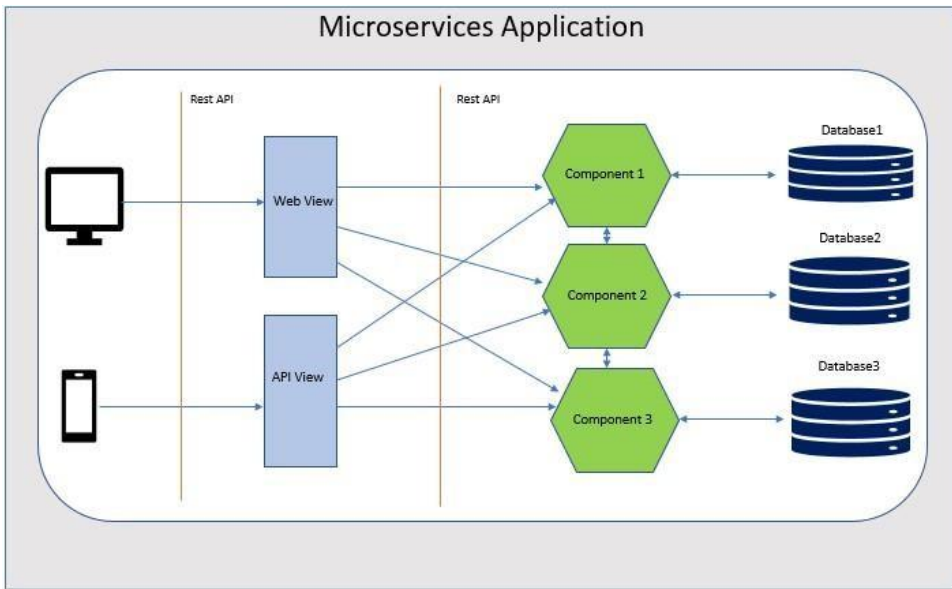


Figure 1.3 Microservice architecture

6.1.5 Real-Time Data Integration via APIs

Given that the system relies on real-time information, it must also integrate with external APIs. For example, the OpenWeatherMap API provides weather, wav forecasts which are necessary for safety and fuel calculations. In addition, the Google Distance Matrix API provides route and distance information. The backend fetches external data points and conveys them through to the mobile app.

The proposed system architecture demonstrates the integration of the mobile frontend, microservices backend, external data sources, and databases, highlighting the offline-first data flow.

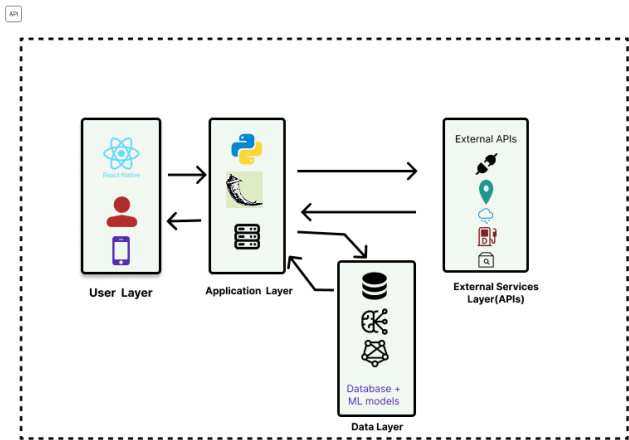


Figure 1.4 Layered way system architecture

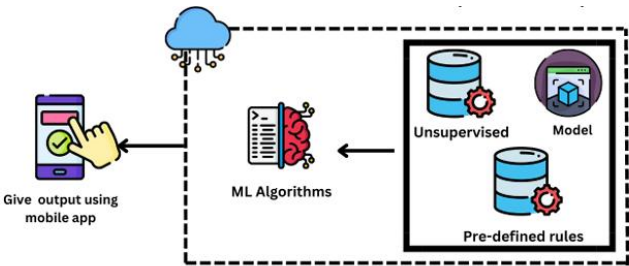


Figure 1.5

According to the figure architecture is structured in four layers. They are presentation layer, application layer, data layer and integration layer. In

Presentation Layer (Mobile App) which is the React Native application providing the user interface. It stores data locally in SQLite for offline access and syncs with the backend when online. Application Layer (Backend Microservices) which is the The FastAPI server, decomposed into independent services (Cost Estimation, Inventory Management, ML Inference, etc.) handling specific business logic. Data Layer which Comprises the cloud PostgreSQL database for permanent storage and the mobile's local SQLite database for offline functionality. Integration Layer which is the external APIs (Weather, Distance, Market Prices) that provide real-time data to the backend services.

The **Presentation Layer** is the React Native mobile application, which provides the main user interface for fishers. It stores data locally in SQLite to support offline use and synchronizes with the backend when internet connectivity is available. The **Application Layer** consists of backend microservices built with FastAPI, each handling independent functions such as cost estimation, inventory management, license validation, and ML inference. The **Data Layer** combines cloud-based PostgreSQL for long-term storage with the mobile's local SQLite database, ensuring both scalability and offline resilience. Finally, the **Integration Layer** connects to external APIs—including weather services, fuel price feeds, market price data, and fish zone information—that supply real-time inputs for cost models and trip planning. Together, these layers allow the system to provide location-aware, real-time decision support that remains usable even in low-connectivity fishing environments

According to Figure 1.4, the machine learning pipeline is structured into four key stages. The process starts with the **External Prediction API**, which collects input variables such as fishing location, distance, gear

usage, fuel consumption, and crew wages. This data then passes through the **Normalization and Feature Extraction layer**, where raw values are cleaned, standardized, and transformed into meaningful features. These processed features send into the **Model Training layer**, where historical trip records are used to build predictive models able of estimating trip costs, catch related profitability, and operational risks. Finally, the **Model Inference Layer** acts as the real-time decision-making engine, integrating trained models into the application to provide instant predictions and actionable insights to fishers while planning their trips

6.1.6. Software Solution

High-quality, priced software can be produced in the smallest amount of time thanks to the systematic process known as SDLC. Producing excellent software that satisfies and surpasses all client requirements and expectations is the aim of SDLC. A comprehensive plan with stages, or phases, that each cover their own process and deliverables is defined and outlined by the SDLC. Following the SDLC reduces project risk and associated costs and speeds up development compared to using alternative production methods.

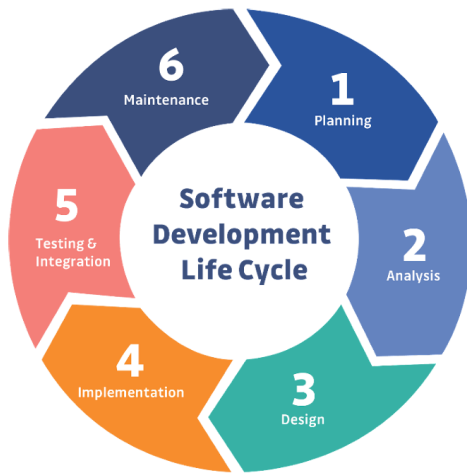


Figure 1.5

6.1.7. Requirement Gathering and Analysis

Fisher Requirements:

- Consultation with fishers and fisheries cooperatives in Sri Lanka to understand pain points in trip planning.
- Analysis of existing informal decision-making processes and cost structures.

- Understanding of literacy and technology adoption levels to inform UI/UX design.

Technical Requirements:

- Offline functionality for core features with seamless background sync.
- Accuracy in cost estimation (>90% correlation with actual trip costs).
- Low-latency response for real-time calculations (<2 seconds).
- Data privacy and security for user financial and operational data.

User Requirements:

- Simple and intuitive screens for entering trip details, viewing predicted costs, and managing inventory
- Support for users with motor impairments, ensuring smooth navigation and interaction on mobile devices.
- Interactive elements and feedback to encourage consistent use and accurate data entry
- Clear display of predicted catch, trip costs, and inventory status for quick understanding.

Business Requirements:

- Integration with existing market price data sources where available.
- Scalability to support a growing user base across multiple regions.
- Low operational cost to ensure long-term sustainability

Feasibility Study

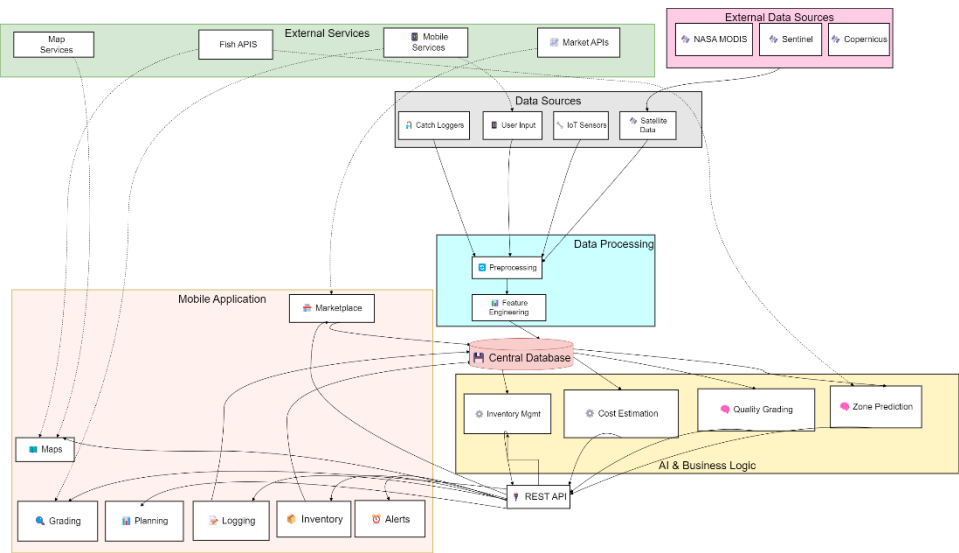
Technical Feasibility: The chosen tech stack (React Native, FastAPI, PostgreSQL) consists of mature, well-documented, and freely available technologies. The research team possesses the required skills in full-stack development, API integration, and data science, making development technically viable.

Financial Feasibility: The project utilizes free-tier cloud services (Render, Supabase), open-source frameworks, and affordable APIs, keeping development and operational costs minimal. The modular design allows for incremental development, reducing financial risk.

Operational Feasibility: The system is built for easy and intuitive use by fishers with different levels of digital literacy. The system is tailored to fit naturally within their pre-trip workflow in order not to disrupt their routines and minimal training will be required for the user. Engagement with local fisheries will provide pathways to pilot and validate the system.

6.1.8 Design Diagrams

6.1.8.1.Overall System Overview Diagram



Our proposed system is based on an integrated, holistic architecture to address the multidimensional problems of Sri Lankan fishermen. The infrastructure is designed in distinct layers that facilitate seamless data flow from collection to decision-making information. Externally sensed satellite data from NASA MODIS, Sentinel provides underlying oceanographic parameters (SST and chlorophyll concentrations), which feed into our processing data layer alongside inputs from IoT sensors, catch loggers, and manual user input. This data undergoes rigorous preprocessing and feature

engineering before being deposited in a centralized database. Business logic and AI layer is the brain of our system, comprising bespoke machine learning models for fish zone prediction (Random Forest/SVM/Neural Networks-based) and fish quality grading (Mobile Net CNN architecture with transfer learning-based), as well as business rules for dynamic trip cost calculation and inventory management. These components communicate through a RESTful API gateway that exposes processed intelligence to a mobile application, enabling features like interactive map-based zone visualization, AI-powered quality grading, trip planning functionality, digital catch logging, marketplace integration, inventory management, and license expiration alerts. The architecture engages in two-way communication with external services like mapping APIs (Google Maps/MapBox), market data feeds, mobile device services, and specialty fisheries APIs, creating an integrated decision-support ecosystem that transforms raw environmental and operational data into actionable intelligence for fishermen at both individual and commercial scales

6.1.8.2. Use Case Diagram

In figure 1.6, the use case diagram visualizes the interactions between the main actor (the Fisher/ user) and the proposed Trip Cost Estimator System. The Fisher begins the process by providing trip detail information such as the type of boat, intended route, and destination. The system then connects to multiple external services to obtain the relevant data. It connects to the Maps & Distance API to receive the expected distance for the trip route, and the Weather API to check for appropriate fishing conditions. Meanwhile, it also accesses available data on fuel, weather, and fish zones from outside sources to ensure that the cost calculations reflect current market rates. The system will also access the internal Inventory Module to check gear usage and availability, while also retrieving the required information from internal Payroll Module Attach on crew wages. When all required items are input, the system collects this data to process for the Total Trip Cost, which the system then displays back to the Fisher. This series of events demonstrates how the system accesses and incorporates both external APIs and internal modules to provide an accurate and reliable trip cost estimation service that assists fishers with their financial and operational decision-making processes.

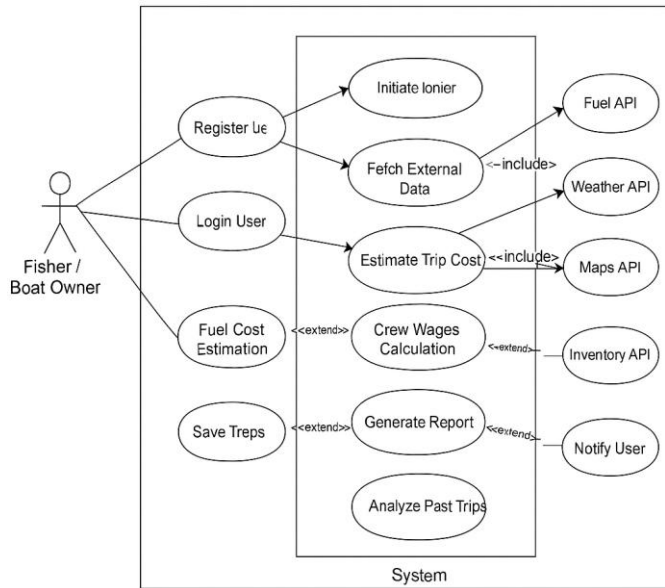


Figure 1.6

6.1.8.3. Sequence Diagrams

In figure 1.7, the sequence diagram shows how a fisherman uses the mobile application to plan a fishing trip and how the various system components make a complete interaction that included cost estimation and decision support. the action starts with the fisherman, who uses the mobile application to start the trip planning function. The mobile application sends a controller message to the Trip Controller (Trip CTRL). The Trip CTRL communicates with the current inventory, fuel, ice, gear, and wages. The Trip CTRL also checks whether the fishing license is valid

and shows alerts if the license will expire soon. The Trip Location module calculates the distance to the selected fishing location and must be provided to the Location API, which includes location information. The Cost Engine uses this input, along with distance, market fuel and ice prices, crew wages, crew wear and tear and updates the cost estimate for the trip. The Validation Module provides license compliance validation. Once the costs and validations are completed, the system updates the inventory status and saves the update into the Data Analysis and Storage module. When completed the fisherman receives a clear decision option and warnings, and advisory, to aid him to make the best decisions before launching on his trip.

Figure 1.7

6.1.9 Dataset Requirements

Trip Data Collection:

- Historical trip data from pilot users (costs, catch, location, crew size).
- Fuel price fluctuations from local sources.
- Fishing zone coordinates and metadata.

Data Preprocessing:

- Cleaning and normalization of historical data.
- Feature engineering for ML model (e.g., creating seasonal indicators, weather flags).
- Geospatial data formatting for API consumption

Validation Data:

- Standardized assessment scores
- Functional outcome measures
- User engagement and satisfaction metrics

6.2.0 Implementation Plan

Our development will be agile and iterative, broken into four clear phases to ensure we build a robust and user-friendly application.

Phase 1: Core Backend & Data Foundation

This phase is about building the brain of the operation. We will set up the server and create the logical rules that power the app.

Server & Database Setup

- We will establish our Python FastAPI server and design the PostgreSQL database schema to securely store user data, trip histories, and fishing zone information

Cost Estimation Engine

- This is the heart of the project. We will develop the core algorithm that calculates trip costs based on dynamic inputs like distance, fuel prices, crew wages, and gear depreciation

Basic API Development: .

- We will build the essential API endpoints that the mobile app will use to communicate with the server, ensuring everything is secure with user authentication

Phase 2: Mobile App & Offline Development

This phase focuses on building the interface that fishers will interact with, ensuring it works flawlessly even without an internet connection.

1. App Framework & UI

- Using React Native, we will build the core screens of the mobile app: the login, dashboard, and the main trip planning form.

2. Offline-First Functionality

- We will implement SQLite database storage directly on the phone. This allows fishers to input data, check their inventory, and view past trips even when they are out at sea with no signal

3. Sync Mechanism

- We will create a background process that automatically synchronizes the locally stored data with the cloud database whenever the phone reconnects to the internet

Phase 3: Advanced Features & Integration (Month 4)

1. External API Integration

- We will connect to the OpenWeatherMap and Google Distance Matrix APIs. This will pull in live weather forecasts and calculate accurate travel distances to chosen fishing zones

2. Machine Learning Model Integration

- We will develop and integrate a basic predictive model. By analyzing historical catch data, this model will provide an estimated catch prediction for the planned trip, which is key to estimating potential revenue.

3. Inventory & Compliance Services.

- We will finalize the features that track gear status and check fishing license validity, setting up automated alerts for missing items or upcoming expiry dates

Phase 4: Testing, Refinement & Launch

The final phase is dedicated to rigorous testing with real users to ensure the app is reliable, accurate, and ready for use.

1. Pilot Testing with Fishers

- We will conduct hands-on testing with a small group of Sri Lankan fishers. This is crucial for getting feedback on usability and real-world accuracy

2. Bug Fixing & Performance Tuning

- Based on pilot feedback, we will fix any issues, refine the user interface for clarity, and optimize the app's performance and battery usage.

3. Deployment

- We will deploy the finalized backend server to a cloud platform and publish the mobile app to the Google Play Store, making it available for downloading

6.2.1 Testing Methodology

We will test the system from three critical angles to ensure it is technically sound, practically useful, and easy to adapt.

Technical Testing:

- Verify the cost estimation accuracy by comparing the app's predictions against actual trip expenses logged by our pilot users.
- Measure the ML model's performance using standard metrics (Precision, Recall) to ensure its predictions are reliable.
- Conduct comprehensive testing of the offline capability to ensure all the essential offline mode functionalities work without an internet connection.
- Check the sync reliability to ensure data is never lost when moving between offline and online states.

Field Testing::

- Run pilot studies with fishers in their actual working environment, observing how they use the app as part of their daily routine.

- Perform a comparative analysis to see if users of the app experience better trip outcomes than those relying on traditional methods.
- Conduct usability sessions with fisheries cooperatives to ensure the app fits well into their ecosystem.

User Experience Testing:

- Ensure the interface is intuitive for users with varying levels of tech experience, using large buttons, clear icons, and minimal text.
- Evaluate the clarity of information, ensuring that profit/loss estimates and alerts are immediately understandable.
- Gather feedback on feature usefulness to understand which parts of the app provide the most value and which need improvement

6.3 Challenges and Mitigation Strategies

Technical Challenges:

- 1 **Challenge:** Unreliable or No Internet Connection.

Mitigation: This is the core reason for our **offline-first architecture**. All essential data is stored on the phone itself, making connectivity an enhancement, not a requirement

- 2 **Challenge:** Accuracy of Predictions.

Mitigation: We will start with a rules-based cost engine. The ML catch prediction will be introduced as an **optional "advisory" feature** first.

Its accuracy will improve over time as more fishers use the app and contribute data.

3 **Challenge** :Data Security for Financial Information.

Mitigation: All data will be encrypted, both on the phone and during transmission. User accounts will be protected with secure authentication.

Implementation Challenges:

- **User Adoption and Digital Literacy:** The design will be simple and icon driven. We will create short video tutorials and work with local community leaders to demonstrate the app's value and build trust
- **Integrating Dynamic Prices:** Initially, fuel and fish prices may need to be manually updated by users or admins. We will design a simple interface for this, with a long-term goal to automate it through API integrations with local authorities.
- **Scalability:** Design a modular architecture to support multiple simultaneous users

6.4 Expected Outcomes and Evaluation Metrics

Technical Outcomes:

- A fully functional, **offline-capable mobile application** published on the app store.
- **>90% accuracy** for the cost estimation engine when compared to real-world trip costs.
- A functional predictive model that provides catch estimates with a reasonable degree of accuracy for initial launch.

Socio-Economic Outcomes:

- A measurable reduction in the number of unprofitable fishing trips among our pilot user group.
- Positive feedback from fishers indicating reduced pre-trip anxiety and more confidence in their planning.
- High user satisfaction (>4/5 rating) in post-pilot surveys

User Experience Outcomes:

- A successful proof-of-concept that demonstrates the value of data-driven decision support for small-scale fisheries.

- A solid foundation for future scalability in other regions and the addition of new features like direct market linkages.

7. PROJECT REQUIREMENTS

7.1 Functional Requirements

The system must successfully estimate trip profitability and manage operational readiness.

- **Real-time Trip Cost Calculation:** Dynamically calculate projected expenses (fuel, ice, crew wages, gear depreciation) based on user input and live data.
- **Integration of External Data:** Automatically pull in real-time fuel prices, weather forecasts, and sea conditions to inform cost and safety calculations.
- **Profit/Loss Forecasting:** Provide a clear pre-trip estimate of potential profit or loss by combining cost projections with ML-based catch predictions.
- **Route Optimization:** Suggest the most fuel-efficient route to the selected fishing zone using geospatial APIs.

The system must track inventory and regulatory compliance.

- **Gear & Inventory Management:** Allow fishers to log their gear (nets, lines, safety equipment) and track their condition, sending alerts for damaged or missing items.
- **License & Permit Validation:** Store license expiry dates and provide proactive alerts to prevent fines or catch confiscation.

- **Pre-Trip Readiness Check:** Run an automated check before trip planning to confirm inventory and license status.

The system must provide actionable insights and historical tracking.

- **Trip History & Analytics:** Maintain a log of all past trips (costs, catch, profit/loss) for review and trend analysis.
- **Visual Data Dashboard:** Display historical performance through charts and graphs to help fishers identify patterns and improve decision-making over time.
- **Offline-First Operation:** Ensure all core functionalities—saving trips, checking inventory, reviewing history—are available without an internet connection

The system must use data-driven methods for prediction.

- **ML-Powered Cost Prediction:** Utilize machine learning models trained on historical and open data to estimate potential cost for a given zone based trip

7.2 Non-Functional Requirements

- Usability: The interface must be simple, intuitive, and usable by fishers with varying levels of literacy and digital experience. It should rely heavily on icons, large buttons, and minimal text.
- Reliability & Accuracy: The cost estimation engine must be highly accurate (>90% correlation with actual costs). The application must be stable and not crash.
- Offline-First Performance: The application must launch and perform all critical functions quickly, even on lower-end Android devices with no internet connection.
- Data Privacy & Security: All user data, especially financial and trip history, must be stored securely and encrypted. User accounts must be protected.
- Scalability: The backend architecture should be able to support a growing number of users from different regions without performance degradation

7.3 Hardware Requirements

- Fisherman's Device: An Android smartphone (version 8.0 or newer) with GPS capabilities. The app will be designed to be lightweight and run on affordable devices.
- Backend Server: A cloud-based virtual machine (e.g., AWS EC2, DigitalOcean Droplet, or free-tier Render/Heroku) to host the Python/FastAPI backend and PostgreSQL database

7.4 Software Requirements

7.4.1 Development Platforms

- Frontend: React Native (Expo) framework for cross-platform mobile development.
- Backend: Python 3.9+ with the FastAPI web framework for building high-performance APIs.
- IDE: Visual Studio Code for development.
- Version Control: Git for source code management

7.4.2 Data, Analytics & Machine Learning

- Data Analysis Libraries: Pandas and NumPy for data manipulation and processing.
- Machine Learning Framework: Scikit-learn for developing and training regression models (e.g., Random Forest, Gradient Boosting) for catch prediction.

- **Data Visualization:** Chart.js or a similar React Native library for displaying analytics and history within the app.

7.4.3 External Service Integrations

- **Maps & Distance API:** Google Distance Matrix API (or a similar alternative) for calculating routes and distances.
- **Weather API:** OpenWeatherMap API (free tier) for fetching marine weather forecasts.
- **Cloud Hosting:** A platform like Render or Heroku for deploying the backend API.
- **Local government fuel price API** (if available) or **PetrolPrices API** (international)
- **Inventory & Supplies API:** Custom API connected to backend DB, optionally linked with ERP/stock APIs

7.4.4 Database and Storage

- **Mobile Database:** SQLite for reliable offline storage on the mobile device.
- **Cloud Database:** PostgreSQL for centralized secure storage of user data and historical records.
- **State Management:** Redux Toolkit with Redux Persist for managing application state and catching data locally.

8. PROJECT BREAKDOWN

8.1 Gantt Chart

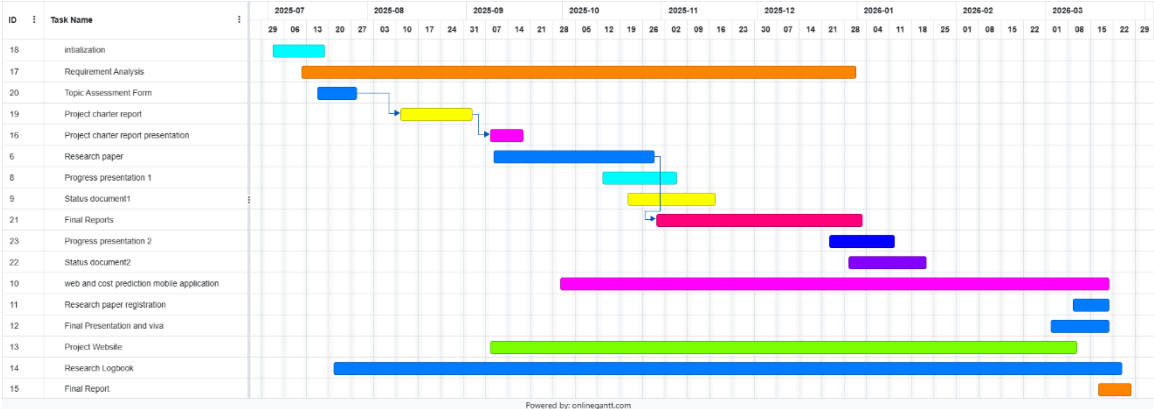


Figure 5.1: Gantt Chart

8.2 Work Breakdown Structure

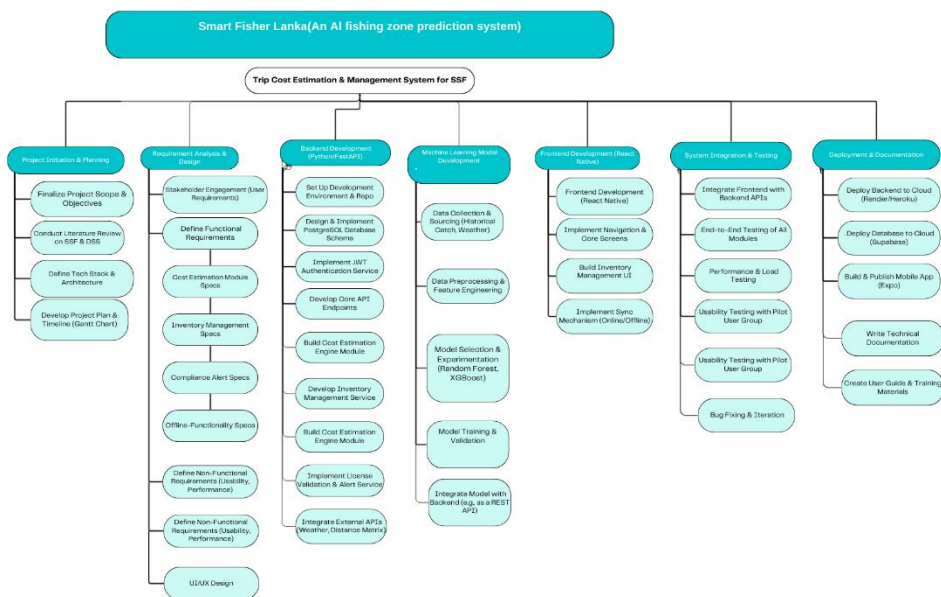


Figure 5.2: Work Breakdown Structure

9. DESCRIPTION OF PERSONAL AND FACILITIES

Facilitators:

- Prof.Pradeep Abeygunawardhena– Sri Lanka Institute of Information Technology
- Mr. Ashvinda Iddamalgoda– Sri Lanka Institute of Information Technology

10. Budget

Task	Cost
Cloud Server (model training, backend & APIs)	\$60 – \$200 / year
Domain Name Registration	\$10 – \$15 / year
API Services (satellite data, maps, vision, notifications, etc.)	\$25 (one-time)
Mobile App Publishing (Google Play Developer Account)	\$50 – \$150
Data Acquisition & Dataset Preparation (satellite, fish images, market data)	\$50 – \$150
IoT Devices & Field Equipment (GPS/catch loggers, testing tools)	\$100 – \$200
Field Testing & Validation (with fishermen, market trials)	\$100 – \$200
Contingency & Miscellaneous	\$15 – \$30
Total (First Year)	\$400 – \$1000

11. COMMERCIALIZATION & SUSTAINABILITY

11.1 Target Audience

- Primary Users: Small-scale fishers in Sri Lanka, specifically boat owners and captains responsible for planning trips and managing finances.
- Secondary Users: Fisheries cooperatives and community organizations that support fisher communities with data-driven insights and management.

11.2 Marketplace Strategy

- Direct Community Engagement: The primary strategy involves partnering with Sri Lankan Fisheries Cooperatives and NGOs working in coastal areas. They act as trusted channels for dissemination, training, and feedback.
- App Stores: The application will be available for download on the Google Play Store, making it accessible to any fisher with an Android smartphone

11.3 Sustainable Funding Model

Instead of subscription fees from fishers, which would be a significant barrier to adoption, the project will explore these sustainable models:

- **Freemium Model for Cooperatives: Basic app features will be free for all individual fishers.** Advanced analytics dashboards for cooperatives (showing aggregate, anonymized data for their entire membership) could be offered under a low-cost premium model (~Rs. 5000/year per cooperative).

- **Grant Funding & NGO Partnerships:** Seek funding from development agencies, NGOs, and government grants focused on **sustainable livelihoods, poverty reduction, and climate resilience** in fishing communities. The tool directly supports these goals.
- **Corporate Social Responsibility (CSR) Sponsorship:** Partner with corporations in Sri Lanka (e.g., in the telecommunications, banking, or insurance sectors) to sponsor the project. Their branding could be tastefully featured within the app as a "Proudly Supported By" message.
- **Donations:** Implement a "Support Our Development" model within the app, allowing users and benefactors to make voluntary donations to support server costs and future development.

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13.APPENDICES



Class Portfolio

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